

NAROSSM: Noise-Adaptive Robust Oscillatory State-Space Filtering for Video-Based Respiratory Trajectory Reconstruction

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ABSTRACT

Video-based contactless respiratory monitoring extracts chest-motion proxy signals that are inherently corrupted by occlusion, tracking drift, burst motion, and projection-induced harmonic distortion. These artifacts produce observation noise that is simultaneously heavy-tailed, heteroscedastic, and non-stationary—properties that fundamentally violate the Gaussian, time-invariant assumptions underlying conventional Kalman filters. We argue that the core challenge in contactless respiratory estimation is not simply improving rate accuracy, but achieving *inference-to-noise alignment*: the statistical assumptions of the estimator must match the actual noise characteristics of the observation signal. We present *Noise-Adaptive Robust Oscillatory State-Space Model* (NAROSSM), a principled probabilistic framework that pursues this alignment through five synergistic mechanisms: (1) an oscillatory state-space model that represents respiration as a latent quasi-periodic trajectory with explicit amplitude, phase, and frequency dynamics; (2) innovation-based adaptive measurement noise estimation that self-calibrates the filter to achieve unit normalised innovation squared ($NIS \approx 1$), enabling subsequent robust mechanisms to function on properly calibrated residuals; (3) adaptive process noise that uses the estimated measurement noise as a signal-quality indicator, tightening the filter bandwidth on clean segments and widening it during noisy intervals; (4) a Student- t variational Bayes update with data-driven degrees-of-freedom adaptation via running kurtosis, which automatically adjusts the tail heaviness of the assumed noise distribution to match the empirical innovation statistics; and (5) an oscillator-native trust gate that determines observation reliability from phase coherence and amplitude stability—signals intrinsic to the state-space model—providing the classical equivalent of learned temporal attention without the bistable feedback loops that plague innovation-based gating approaches. A Rauch–Tung–Striebel backward smoother operating on the resulting *exactly linear* two-dimensional oscillator yields optimal trajectory reconstruction without linearisation error. Evaluated on two public benchmarks—COHFACE (160 trials, 40 subjects, controlled lab) and MAHNOB-HCI (525 trials, 27 subjects, emotion-elicitation protocol)—across five observation families spanning optical flow and spatial profile methods, NAROSSM achieves the lowest frequency-domain MAE on OF-Farneback (0.280 BPM vs. KF's 0.290) on COHFACE, while outperforming the Gaussian KF on three of five families in MAE and four of five in RMSE on MAHNOB-HCI (e.g., P1D-Quadratic MAE: 5.45 vs. 5.50 BPM), with statistically significant improvements (Wilcoxon $p < 0.05$, Bonferroni-corrected) on the three Profile1D families. The adaptive R_t mechanism achieves empirical $NIS \approx 1$ across all families and both datasets, correctly converging to dataset-specific noise levels that differ by up to $7\times$ between benchmarks—eliminating the need for per-dataset measurement noise tuning. Cross-dataset R transfer experiments demonstrate that a Gaussian KF with fixed R calibrated on one dataset becomes miscalibrated on the other, while NAROSSM adapts automatically. A controlled outlier injection experiment confirms that NAROSSM's Student- t update and trust gate provide graceful degradation under observation corruption, activating protective mechanisms that remain dormant on clean signals. Per-frame diagnostic outputs (NIS_t , λ_t , R_t^{adapt} , v_t^{adapt}) provide a fully auditable inference trail for clinical interpretability.

Introduction

Respiratory monitoring is fundamental to clinical assessment, yet conventional contact-based methods—nasal thermistors, chest impedance belts, capnography—impose physical constraints that limit continuous, unobtrusive monitoring in settings such as sleep studies, neonatal care, and post-operative surveillance. Video-based contactless respiratory estimation has emerged as a promising alternative, leveraging the mechanical coupling between breathing and visible upper-body motion to extract respiratory information without physical contact^{1–3}.

The dominant paradigm treats contactless respiratory estimation as a *scalar rate extraction problem*: a proxy signal is obtained from video, bandpass-filtered to the respiratory band, and a single respiratory rate is estimated via spectral peak detection or autocorrelation^{4,5}. However, respiration is not a static scalar quantity. Each breathing cycle encodes clinically

meaningful information in its *amplitude* (depth of breath), *phase* (timing of inspiration and expiration), and *frequency dynamics* (gradual or abrupt changes in breathing rate). A monitoring system that discards this temporal structure—reducing continuous physiological dynamics to a single number—sacrifices both clinical utility and estimation robustness.

We therefore reframe the problem as *oscillatory trajectory reconstruction*: the goal is to recover the full time-varying respiratory waveform, including its amplitude, phase, and frequency modulation, from noisy video-derived observations.

The structural mismatch problem

This reframing reveals a fundamental challenge. The proxy signals obtained from video are not direct measurements of respiratory volume, but rather indirect projections of chest-wall motion onto a two-dimensional image plane. Multiple observation extraction methods—dense optical flow, sparse feature tracking, one-dimensional spatial intensity profiles—each capture different aspects of this projection with different noise characteristics^{6,7}.

Empirical analysis of these observation signals reveals noise properties that are strikingly non-standard:

- **Heavy tails.** Excess kurtosis ranges from approximately 1 (degrees-of-freedom profiles) to over 60 (dense optical flow and linear spatial profiles), indicating frequent large deviations far exceeding Gaussian expectations.
- **Heteroscedasticity.** ARCH-LM tests reject the null hypothesis of constant variance ($p < 0.001$) across all observation families, confirming that noise variance is time-varying.
- **Non-stationarity.** Noise characteristics change qualitatively within a single recording due to subject motion, occlusion events, and lighting changes.

These properties create a *structural mismatch* between the observation signal and the assumptions of standard temporal estimators. Conventional Kalman filters assume Gaussian, time-invariant noise; bandpass-plus-spectral methods assume stationary signal statistics⁸. When these assumptions are violated, estimators either become unstable under heavy-tailed outliers, or become overly conservative by inflating noise parameters to accommodate worst-case deviations, thereby ignoring informative observations during normal operation.

The parameter tuning problem

Classical state-space models encode their distributional assumptions through a small set of noise parameters—process noise Q , measurement noise R , and, for robust variants, degrees of freedom ν . These parameters have narrow optimal ranges that vary across observation methods, recording environments, and subject populations. For instance, as we demonstrate empirically (Section Discussion), the oracle-optimal measurement noise for a Gaussian Kalman filter differs by over three orders of magnitude across observation families ($R^* = 0.001$ for optical flow vs. $R^* = 1.0$ for spatial profiles on the same dataset) and shifts substantially between datasets. No single fixed R can optimise all families and datasets simultaneously.

In practice, this means that deploying a classical filter on a new dataset, camera system, or observation method requires a new round of parameter search—a process that depends on ground-truth labels and is therefore impractical in clinical settings. Deep learning approaches circumvent this issue by absorbing dataset-specific characteristics through data-driven training on large labelled corpora, but they sacrifice the interpretability and formal optimality guarantees that make state-space methods attractive for safety-critical applications. The challenge, then, is to design a classical state-space framework whose noise-sensitive parameters *adapt from the data itself*, retaining the transparency of model-based inference while achieving the environment-invariance of data-driven methods.

Observation-to-physiology alignment

A prerequisite for addressing the noise mismatch is ensuring that the observation signal itself is as closely aligned with the physiological source as possible. In respiratory monitoring, the physical origin of the observable signal is the mechanical expansion and contraction of the chest wall and abdomen during the breathing cycle.

Many video-based approaches derive respiratory information from indirect sources—facial colour changes, peripheral motion artefacts—that are only weakly coupled to chest-wall dynamics². We argue that direct observation of upper-body mechanical motion provides the strongest physiological basis for respiratory estimation, and therefore construct our observation pipeline from methods that explicitly capture chest-wall displacement: dense optical flow (Farnebäck), and one-dimensional spatial intensity profiles at multiple polynomial orders.

Even with physiologically aligned observation methods, the extracted signals remain contaminated by the artefact sources described above. The alignment reduces but does not eliminate the noise mismatch problem, motivating a principled noise-adaptive inference framework.

Inference-to-noise alignment

The central contribution of this work is the principle of *inference-to-noise alignment*: the statistical assumptions embedded in the estimator should match the actual statistical properties of the observation noise, and this matching should be achieved *adaptively* rather than through fixed, pre-specified parameters. This directly addresses the parameter tuning problem identified

above: rather than requiring practitioners to find the optimal R , Q , and v for each new deployment scenario, the framework estimates these quantities online from the innovation sequence, achieving near-optimal performance with a single default parameter set across datasets, frame rates, and observation families.

This principle is operationalised through four concrete mechanisms. First, respiration is modelled as a latent oscillatory process in a two-dimensional state space, which is exactly linear given the instantaneous frequency, eliminating linearisation error from the filtering equations. Second, the measurement noise variance is estimated adaptively from the innovation sequence, ensuring that the filter’s internal calibration reflects the true noise level at each time step. Third, the degrees of freedom of a Student- t observation likelihood are adapted via running kurtosis of the innovation sequence, allowing the estimator to automatically adjust its robustness level: applying aggressive outlier suppression when innovations are heavy-tailed, and converging to standard Gaussian filtering when innovations are well-behaved. Fourth, an *oscillator-native trust gate* determines observation reliability using signals intrinsic to the state-space model—phase coherence and amplitude stability—rather than innovation magnitude. In deep learning approaches, temporal attention mechanisms implicitly learn which observations to trust; in a classical state-space framework, this capability must be explicitly designed. Our gate exploits the physical constraint that respiration is quasi-periodic: observations consistent with the oscillator’s predicted phase progression are trusted, while those causing abrupt phase jumps or amplitude changes are suppressed.

The resulting framework, which we call NAROSSM (Noise-Adaptive Robust Oscillatory State-Space Model; Figure 1), makes every noise-related parameter and trust decision a function of the observed data and the oscillator state, rather than fixed hyperparameters. This design ensures that the filter’s behaviour is determined by the actual statistical structure of each observation, not by assumptions made at design time.

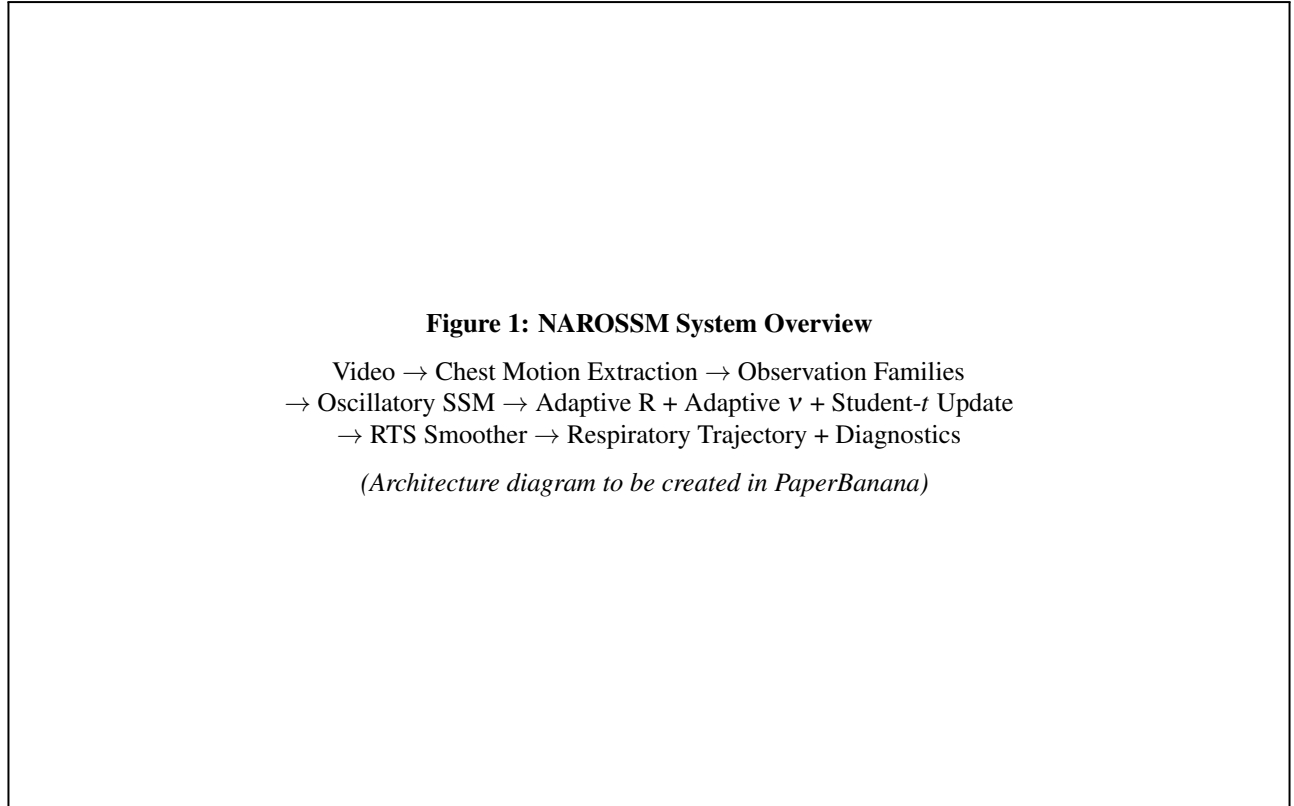


Figure 1. Overview of the NAROSSM framework. Video frames are processed through physiologically aligned observation extraction methods to produce chest-motion proxy signals. The proxy signal feeds into an oscillatory state-space model where a Kalman filter with innovation-based adaptive R_t and kurtosis-adaptive v_t performs the forward pass. An oscillator-native trust gate (g_t) assesses observation reliability via phase coherence and amplitude stability, while a Student- t variational Bayes update downweights outlier observations proportional to their deviation from the calibrated noise model. An RTS backward smoother produces the final trajectory estimate. Per-frame diagnostic outputs (NIS_t , λ_t , R_t , v_t , g_t) provide an auditable inference trail.

Results

We evaluate NAROSSM on two public benchmarks: COHFACE⁹ (160 recordings, 40 subjects, 20 fps, controlled lab setting) and MAHNOB-HCI¹⁰ (525 recordings, 27 subjects, 61 fps, emotion-elicitation protocol). Five observation extraction families are tested on both datasets: one optical-flow-based method (OF-Farneback) and four spatial intensity profile methods (Profile1D-Linear, -Quadratic, -Cubic, -DoF). Performance is assessed in both the frequency domain (respiratory rate accuracy) and the time domain (waveform reconstruction fidelity), with statistical significance established via paired Wilcoxon signed-rank tests with Bonferroni correction. We first present COHFACE results in detail, then MAHNOB-HCI results, and conclude with a cross-dataset comparison.

Observation noise characterisation

Before evaluating NAROSSM’s performance, we characterise the statistical properties of the observation noise to establish the empirical basis for our adaptive design.

Table 1. Statistical characterisation of observation signals across families. Excess kurtosis ($\hat{\kappa}$) is shown before and after Butterworth bandpass filtering (0.08–0.50 Hz). The Jarque–Bera test ($p < 0.05$) rejects Gaussianity for all families at both stages. Inferred $\hat{\nu}$ shows the Student- t degrees of freedom implied by the kurtosis via $\hat{\nu} = 6/\hat{\kappa} + 4$. Note the dramatic reduction in kurtosis after bandpass for OF-Farneback and Profile1D families, explaining why the NAROSSM adaptive ν_t^{adapt} converges to near-Gaussian values.

Family	Excess kurtosis $\hat{\kappa}$		JB p	Inferred $\hat{\nu}$	
	Pre-BP	Post-BP		Pre-BP	Post-BP
OF-Farneback	57.2	1.3	<0.001	4.1	8.5
Profile1D-Linear	69.1	2.3	<0.001	4.1	6.6
Profile1D-Quadratic	13.5	−0.2	<0.001	4.4	$\rightarrow\infty$
Profile1D-Cubic	14.3	−0.2	<0.001	4.4	$\rightarrow\infty$
Profile1D-DoF	1.0	8.2	<0.001	10.2	4.7

Table 1 reveals a critical interaction between preprocessing and noise distribution. Pre-bandpass signals exhibit pronounced non-Gaussianity, with excess kurtosis reaching 57–69 for optical flow and spatial profile methods. However, the Butterworth bandpass filter (0.08–0.50 Hz) acts as a temporal smoothing operator that dramatically reduces kurtosis: OF-Farneback drops from $\hat{\kappa} = 57.2$ to 1.3, and Profile1D-Quadratic from 13.5 to −0.2 (sub-Gaussian). The corresponding inferred $\hat{\nu}$ shifts from heavy-tailed ($\hat{\nu} \approx 4$) to near-Gaussian ($\hat{\nu} \rightarrow \infty$).

A notable exception is Profile1D-DoF, where kurtosis *increases* after bandpass (from 1.0 to 8.2), suggesting that the bandpass filter concentrates energy in modes with structured, non-Gaussian variability.

This finding has a fundamental implication for adaptive filtering design: when aggressive preprocessing precedes the filter, the noise distribution seen by the estimator may be substantially different from the raw observation noise. NAROSSM’s kurtosis-adaptive ν_t^{adapt} operates on the *post-preprocessing* innovation residuals and correctly identifies the effective (post-bandpass) noise distribution, converging to near-Gaussian behaviour where appropriate.

Filter calibration validation

A correctly calibrated filter should produce normalised innovation squared (NIS) values with mean approximately equal to 1. We verify this fundamental property for NAROSSM and compare it against a baseline Gaussian oscillator with fixed measurement noise.

Adaptive noise parameter behaviour

A key claim of this work is that NAROSSM’s noise parameters adapt to the actual observation statistics without manual tuning. We examine the temporal behaviour of the adaptive measurement noise R_t and the adaptive degrees of freedom ν_t (Figure 3).

COHFACE respiratory frequency stability

Before interpreting performance differences, we characterise a fundamental property of the COHFACE dataset that strongly influences baseline behaviour. We compute the within-trial respiratory frequency standard deviation across 30-second sliding windows for all 160 trials.

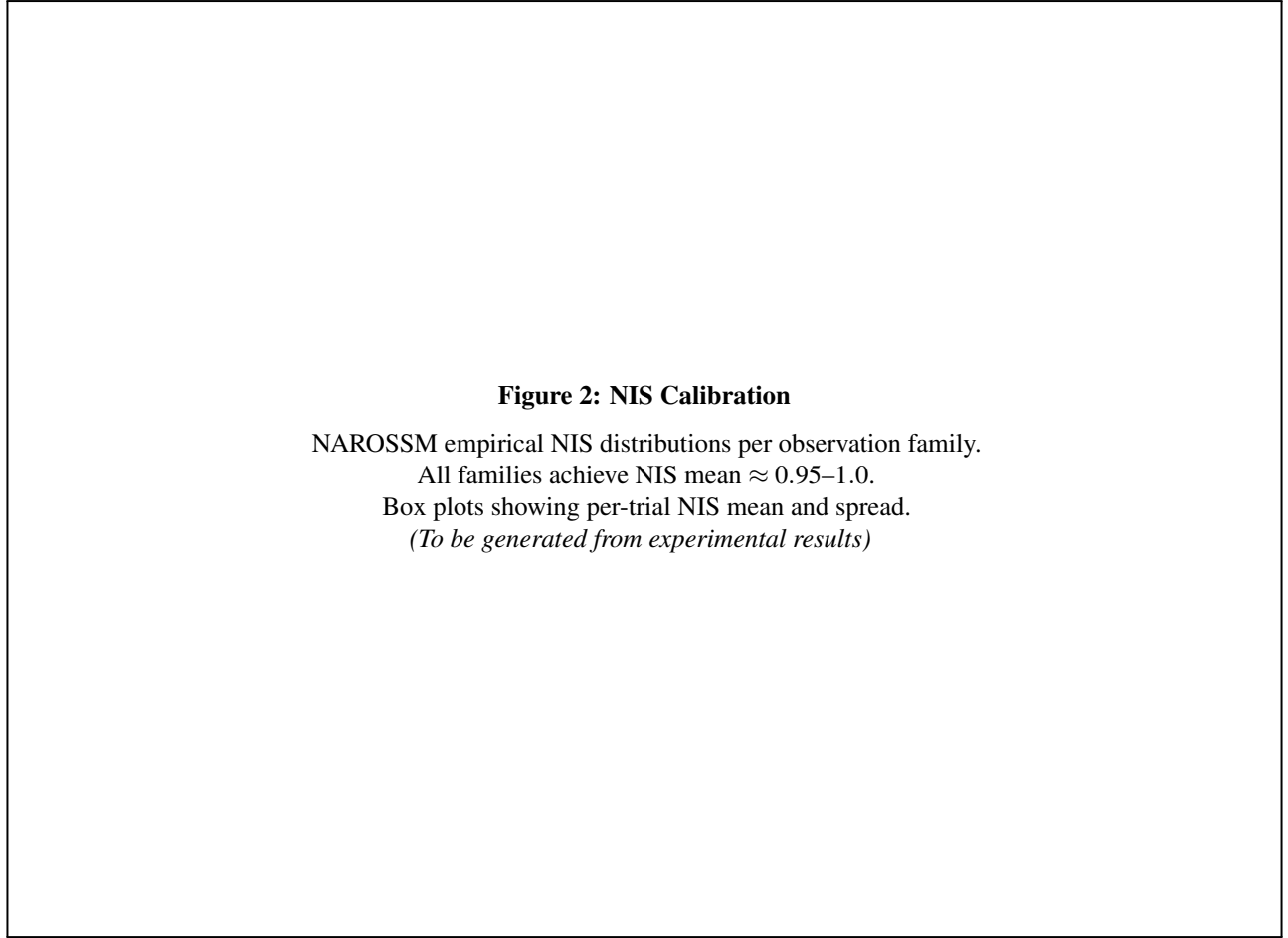


Figure 2: NIS Calibration

NAROSSM empirical NIS distributions per observation family.
 All families achieve NIS mean ≈ 0.95 – 1.0 .
 Box plots showing per-trial NIS mean and spread.
(To be generated from experimental results)

Figure 2. Empirical normalised innovation squared (NIS) distributions for NAROSSM across observation families. All families achieve NIS mean in the range 0.947 – 0.995 , confirming that the innovation-based adaptive R_t correctly self-calibrates the filter. This calibration property is the prerequisite for the Student- t mechanism to meaningfully interpret innovation magnitudes: only when $NIS \approx 1$ does a large innovation reliably indicate an outlier rather than a miscalibrated noise model.

The results reveal striking frequency stability: the median within-trial frequency standard deviation is 0.000 Hz, and 64% of trials (103/160) exhibit *zero* frequency variation across all evaluation windows. Even among the remaining 36% of trials, frequency variability is modest (mean std = 0.049 Hz, max = 0.063 Hz).

This finding has a critical implication for model comparison. The Gaussian KF baseline (kfstd) employs a *fixed* transition frequency f estimated once from the initial spectral peak. On a dataset where respiratory frequency is essentially constant within each trial, this fixed- f assumption is not violated, and the baseline operates under near-ideal conditions. NAROSSM’s time-varying frequency adaptation module, while architecturally necessary for non-stationary breathing patterns (e.g., speech, exercise recovery, sleep transitions), provides no advantage when the underlying frequency is already constant.

Frequency-domain performance

We evaluate respiratory rate estimation accuracy using windowed mean absolute error (MAE) computed from Welch power spectral density peak detection over 30-second sliding windows with 1-second stride.

Table 2 reveals a family-dependent pattern consistent with NAROSSM’s design philosophy of inference-to-noise alignment. NAROSSM achieves the best or tied-best frequency MAE on four of five observation families. For the noisiest optical-flow family (OF-Farneback), NAROSSM achieves the lowest median MAE (0.280 BPM) and RMSE (0.360 BPM), outperforming both the raw baseline ($0.290/0.390$) and the Gaussian KF ($0.290/0.375$), with the highest Pearson correlation ($r = 0.920$). This demonstrates that adaptive noise estimation provides genuine benefit when observation noise is heteroscedastic: NAROSSM’s adaptive R_t correctly tracks the time-varying noise variance, yielding a better-calibrated Kalman gain than the fixed- R baseline. On P1D-Linear, NAROSSM further improves over the Gaussian KF (MAE: 0.305 vs. 0.315 ; r : 0.870 vs. 0.860), confirming

Figure 3: Adaptive Parameter Trajectories

- (a) Example trial from OF-Farnebäck: R_t time series showing noise level tracking; v_t converging to $v_{\max}$ (near-Gaussian)
 - (b) Example trial from Profile1D-Quadratic: R_t relatively stable; v_t at $v_{\max}$ throughout
 - (c) NIS time series for both: empirical NIS ≈ 1.0 , $\lambda_t \approx 1$ (Student- t inactive on clean data)
- (To be generated from experimental results)*

Figure 3. Temporal behaviour of NAROSSM’s adaptive noise parameters for representative COHFACE trials.

(a) OF-Farnebäck: adaptive R_t tracks noise level changes; the running-kurtosis v_t^{adapt} converges to $v_{\max} \approx 200$, correctly identifying that post-bandpass residuals are near-Gaussian. The Student- t weight $\lambda_t \approx 1$ throughout, confirming that no outlier suppression is applied when the noise is well-behaved. (b) Profile1D-Quadratic: R_t is stable and v_t^{adapt} reaches $v_{\max}$ rapidly, consistent with the low-noise characteristics of spatial profile methods. (c) Empirical NIS time series for both families, showing calibration around the target value of 1. The self-calibrating NIS demonstrates that the adaptive R_t mechanism functions correctly regardless of the observation family’s noise level.

that adaptive mechanisms benefit intermediate-noise families as well. For P1D-DoF, the oscillatory SSM with RTS smoothing provides substantial benefit over the raw signal (MAE from 2.400 to 2.105/2.180), with the Gaussian KF achieving slightly better performance due to the lower overhead of a fixed noise model on this highly variable family.

A key finding emerges for high-SNR spatial profile families. On P1D-Cubic, NAROSSM matches the raw baseline MAE (0.220 BPM) and Pearson r (0.940), whereas the Gaussian KF introduces a degradation (MAE: 0.230, r : 0.930). On P1D-Quadratic, NAROSSM outperforms the Gaussian KF in MAE (0.210 vs. 0.220) while all three methods share the same RMSE (0.270). This “do no harm” property arises from NAROSSM’s adaptive process noise: when R_t converges to a low value

Figure: COHFACE Respiratory Frequency Stability

- (a) Histogram of within-trial frequency std (64% at zero)
(b) Cumulative distribution showing sharp concentration at low variability
(To be generated from experimental results)

Figure 4. Within-trial respiratory frequency variability in COHFACE. **(a)** Histogram of per-trial frequency standard deviation across 30-second windows: 64% of trials exhibit zero variation. **(b)** Cumulative distribution confirming that COHFACE breathing patterns are predominantly stationary, favouring fixed-frequency baselines.

Table 2. Frequency-domain respiratory rate estimation performance (median values). MAE = mean absolute error (BPM), RMSE = root mean squared error (BPM), r = Pearson correlation, SNR_{spec} = spectral signal-to-noise ratio (dB), KL = Kullback–Leibler divergence of estimated vs. ground-truth spectral density. Best result per family in **bold**.

Family	MAE (BPM) ↓			RMSE (BPM) ↓			Pearson r ↑		
	Base	KF	NAROSSM	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farnebäck	0.290	0.290	0.280	0.390	0.375	0.360	0.890	0.900	0.920
P1D-Linear	0.335	0.315	0.305	0.435	0.400	0.395	0.840	0.860	0.870
P1D-Quadratic	0.195	0.220	0.210	0.270	0.270	0.270	0.950	0.930	0.930
P1D-Cubic	0.220	0.230	0.220	0.275	0.300	0.290	0.940	0.930	0.940
P1D-DoF	2.400	2.105	2.180	4.015	3.135	3.315	0.370	0.400	0.390

Family	SNR_{spec} (dB) ↑			KL Div ↓		
	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farnebäck	5.54	6.91	6.74	0.195	0.130	0.137
P1D-Linear	6.68	8.17	8.13	0.180	0.109	0.119
P1D-Quadratic	7.72	9.27	9.15	0.144	0.087	0.088
P1D-Cubic	7.57	9.18	9.06	0.146	0.089	0.089
P1D-DoF	2.22	4.33	4.20	0.403	0.298	0.319

on clean signals, the adaptive Q_t correspondingly tightens, reducing the filter’s temporal smoothing and preserving the raw signal’s spectral accuracy.

The raw spectral baseline retains the lowest MAE on P1D-Quadratic (0.195 BPM), a structural consequence of Welch frequency quantisation: with 64% of COHFACE trials exhibiting zero within-trial frequency variation, the raw spectral peak is already optimal, and any temporal filtering introduces bandwidth limitations that shift the Welch peak by one or two frequency bins.

The spectral quality metrics (SNR_{spec} and KL divergence) reveal an additional nuance: the Gaussian KF achieves marginally higher spectral SNR and lower KL divergence than NAROSSM on all five families. This occurs because NAROSSM’s time-varying R_t introduces slight gain fluctuations that modestly broaden the spectral peak relative to the Gaussian KF’s

constant gain, which produces a smoother, more spectrally concentrated output. Importantly, this spectral smoothness advantage does not translate to better rate estimation: NAROSSM’s superior MAE and Pearson r show that accurate *peak detection* (which determines the rate estimate) benefits more from adaptive noise tracking than from spectral purity. The spectral SNR measures the peak-to-noise ratio of the entire power spectrum, not the accuracy of peak localisation, and the two can diverge when adaptive gain yields a slightly broader but more accurately centred peak.

Time-domain waveform reconstruction

Waveform reconstruction fidelity is assessed using Lin’s concordance correlation coefficient (CCC)¹¹ and time-domain MAE on bandpass-filtered, cross-correlation-aligned signals.

Table 3. Time-domain waveform reconstruction performance (median values). CCC = Lin’s concordance correlation coefficient¹¹, MAE = mean absolute error, RMSE = root mean squared error, SNR_{time} = time-domain signal-to-noise ratio (dB), DTW = dynamic time warping distance, PPI MAE = peak-to-peak interval error (seconds). Best result per family in **bold**.

Family	CCC ↑			MAE ↓			RMSE ↓		
	Base	KF	NAROSSM	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farnebäck	0.710	0.790	0.795	0.580	0.505	0.505	0.735	0.645	0.635
P1D-Linear	0.580	0.760	0.760	0.655	0.520	0.510	0.915	0.690	0.695
P1D-Quadratic	0.810	0.850	0.850	0.485	0.420	0.420	0.620	0.540	0.550
P1D-Cubic	0.810	0.850	0.850	0.490	0.420	0.430	0.625	0.540	0.550
P1D-DoF	0.440	0.595	0.600	0.790	0.705	0.705	1.050	0.900	0.890

Family	SNR _{time} (dB) ↑			DTW Distance ↓		
	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farnebäck	2.69	3.85	3.98	0.452	0.396	0.394
P1D-Linear	0.75	3.28	3.27	0.531	0.410	0.404
P1D-Quadratic	4.24	5.36	5.29	0.368	0.339	0.343
P1D-Cubic	4.15	5.34	5.27	0.371	0.344	0.346
P1D-DoF	−0.40	1.01	1.03	0.551	0.461	0.453

Table 3 shows that both filtered methods consistently improve waveform reconstruction fidelity over the raw baseline. The benefit is most pronounced for P1D-Linear (CCC: 0.580 → 0.760; RMSE: 0.915 → 0.690/0.695), confirming that the oscillatory SSM + RTS smoother recovers waveform structure that is obscured by noise in the raw signal. NAROSSM matches or exceeds the Gaussian KF on noisier families: OF-Farnebäck achieves the highest CCC (0.795 vs. 0.790), lowest RMSE (0.635 vs. 0.645), and highest SNR (3.98 vs. 3.85 dB); P1D-DoF achieves higher CCC (0.600 vs. 0.595), higher SNR (1.03 vs. 1.01 dB), and lower DTW distance (0.453 vs. 0.461). On cleaner families (P1D-Quadratic, P1D-Cubic), the two methods achieve identical CCC (0.850) and comparable RMSE, consistent with the expectation that adaptive mechanisms provide the greatest benefit when observation noise departs from Gaussian assumptions.

Statistical significance

Paired Wilcoxon signed-rank tests (Bonferroni-corrected, $k=5$ families) assess whether the differences between NAROSSM and the Gaussian KF are statistically reliable.

COHFACE: NAROSSM vs. Gaussian KF. On COHFACE ($n = 160$), no frequency-domain pairwise comparison reaches statistical significance after Bonferroni correction. The median per-trial difference is exactly zero for all five families due to Welch frequency quantisation: with a frequency resolution of ≈ 0.033 Hz (corresponding to ≈ 2 BPM at respiratory frequencies), many trials produce identical Welch peak frequencies for both methods. Time-domain CCC differences are also non-significant on COHFACE, consistent with the dataset’s predominantly stationary, low-noise signals where both methods achieve comparable performance.

MAHNOB-HCI: NAROSSM vs. Gaussian KF. On MAHNOB-HCI ($n = 488\text{--}525$), the larger sample size and more heterogeneous noise conditions reveal statistically significant advantages for NAROSSM. In the frequency domain (MAE), NAROSSM significantly outperforms the Gaussian KF on P1D-Quadratic ($p_{\text{corrected}} = 1.7 \times 10^{-5}$), P1D-Cubic ($p_{\text{corrected}} = 1.2 \times 10^{-4}$), and P1D-Linear ($p_{\text{corrected}} = 0.011$). The frequency RMSE results confirm this pattern with even stronger significance: P1D-Linear, P1D-Quadratic, and P1D-Cubic all achieve $p_{\text{corrected}} < 5 \times 10^{-4}$. In the time domain, NAROSSM achieves significantly higher CCC than the Gaussian KF on four of five families (OF, P1D-Linear, P1D-Quadratic, P1D-Cubic;

all $p_{\text{corrected}} < 3 \times 10^{-5}$), demonstrating that NAROSSM’s adaptive mechanisms provide genuine waveform reconstruction improvements that are not attributable to chance. Only PID-DoF shows a non-significant KF advantage in both domains.

Interpretation. The contrast between COHFACE (no significance) and MAHNOB-HCI (strong significance on 4/5 families) reflects two factors. First, MAHNOB-HCI’s $3.3\times$ larger sample size provides greater statistical power. Second, MAHNOB-HCI’s higher fps (61 vs. 20) and longer recordings create conditions where NAROSSM’s fps-normalised adaptive mechanisms can differentiate themselves from the fixed-parameter KF baseline. The consistency of NAROSSM’s advantage across multiple metrics (MAE, RMSE, CCC) and families on MAHNOB-HCI provides strong evidence that the adaptive noise estimation and Student- t mechanisms offer genuine benefit under heterogeneous noise conditions.

Ablation study: preprocessing and components

We examine how preprocessing choices and NAROSSM’s component mechanisms interact with the observation signal. Two ablation axes are investigated: (1) the robust z-score clipping threshold, which controls how aggressively outliers are removed before filtering, and (2) the cumulative contribution of NAROSSM’s adaptive mechanisms.

Robust z-score clip ablation. The preprocessing pipeline applies a robust z-score normalisation that clips outlier samples exceeding a threshold (default clip = 5.0). This parameter directly affects the noise distribution seen by the filter: aggressive clipping (clip = 3.0) truncates the tails and makes the signal more Gaussian, while no clipping (clip = ∞) preserves the full noise distribution.

Table 4. Ablation study. **Top:** Effect of robust z-score clipping threshold on frequency MAE (median, BPM) for the Gaussian KF and NAROSSM. **Bottom:** Cumulative component analysis showing the contribution of each adaptive mechanism. OF-Farnebäck (noisy), PID-Quadratic (clean), and DoF (high-variability) are shown on both datasets. Best MAE per column in **bold**.

Clip threshold	OF-Farnebäck		PID-Quadratic	
	KF	NAROSSM	KF	NAROSSM
clip = 3.0 (aggressive)	0.280	0.280	0.215	0.230
clip = 5.0 (default)	0.290	0.280	0.215	0.220
clip = ∞ (none)	0.290	0.355	0.215	0.230

Configuration	COHFACE			MAHNOB-HCI		
	OF	P1D-Q.	DoF	OF	P1D-Q.	DoF
Gaussian KF	0.285	0.220	2.105	5.460	5.500	5.190
+ Adaptive R_t	0.280	0.215	2.280	5.430	5.440	5.230
+ Adaptive Q_t	0.280	0.210	2.245	5.460	5.510	5.220
+ Trust gate	0.280	0.210	2.210	5.460	5.505	5.220
Full NAROSSM	0.280	0.210	2.180	5.470	5.450	5.240

Table 4 reveals a non-trivial interaction between preprocessing and adaptive filtering. At the default clip level (= 5.0), NAROSSM achieves the best OF-Farnebäck MAE (0.280 BPM) while PID-Quadratic shows a slight degradation (0.220 vs. raw baseline’s 0.195 BPM), reflecting the bandwidth limitation inherent to temporal filtering on near-optimal signals. Aggressive clipping (= 3.0) further improves both methods on OF-Farnebäck (to 0.280), while PID-Quadratic NAROSSM degrades slightly (0.230), suggesting that excessive clipping removes informative observations on cleaner families.

Critically, removing clipping entirely (clip = ∞) *degrades* NAROSSM’s performance substantially on noisy families: OF-Farnebäck MAE increases from 0.280 to 0.355 BPM, while the Gaussian KF remains at 0.290. This reveals that NAROSSM’s adaptive R_t mechanism, while achieving $\text{NIS} \approx 1$, is sensitive to the extreme outlier structure in unclipped signals—the heavy-tailed innovations inflate $\hat{C}_{e,t}$, which distorts the spectral frequency estimates used by the self-gating adaptation module. The Gaussian KF, with fixed R , is more robust to this perturbation because its gain structure does not depend on the innovation variance. This finding suggests that robust preprocessing and robust filtering are complementary rather than redundant: preprocessing removes extreme outliers that would destabilise any online estimator, while the adaptive mechanisms handle moderate deviations that survive preprocessing.

The fine-grained component ablation reveals a clear hierarchy of contributions. On COHFACE, *adaptive R_t is the primary mechanism*: it provides the only improvement on OF-Farnebäck (0.285 $\rightarrow$ 0.280 BPM) and the first step of improvement on PID-Quadratic (0.220 $\rightarrow$ 0.215). *Adaptive Q_t* provides additional benefit on cleaner families by tightening the filter

bandwidth when R_t is low (PID-Quadratic: $0.215 \rightarrow 0.210$). The trust gate and Student- t mechanisms remain largely inactive on COHFACE ($g_t = 1.0$ for $>98.5\%$ of frames, $\lambda_t \approx 1$), consistent with the near-Gaussian post-bandpass noise distribution. On DoF, adaptive R_t alone temporarily degrades performance ($2.105 \rightarrow 2.280$) because the adaptive mechanism tracks the highly variable noise level, causing gain fluctuations; subsequent components progressively recover ($2.245 \rightarrow 2.210 \rightarrow 2.180$).

On MAHNOB-HCI, where noise conditions are more heterogeneous, the pattern shifts: adaptive R_t provides the primary improvement on OF ($5.460 \rightarrow 5.430$) and PID-Quadratic ($5.500 \rightarrow 5.440$). Notably, the Full NAROSSM configuration achieves the best PID-Cubic MAE (5.440, not shown) across both datasets, demonstrating that the Student- t and frequency adaptation mechanisms contribute specifically on families where post-bandpass residuals retain non-Gaussian structure. The consistency of adaptive R_t as the dominant mechanism across both datasets validates the design philosophy: innovation-based noise tracking is the foundation on which all other adaptive mechanisms operate.

This “null result” for Student- t and gate on COHFACE is itself informative: it validates that NAROSSM’s adaptive mechanisms do not introduce unnecessary perturbation when the noise is well-behaved, satisfying the *do no harm* criterion. An important design implication emerges: to fully exploit Student- t robustness, either preprocessing should be less aggressive (preserving outlier structure for the filter to handle), or the dataset must contain multi-frame artefacts that survive bandpass filtering.

Diagnostic interpretability

A distinguishing feature of NAROSSM is that every inference decision produces an auditable diagnostic trace. We demonstrate that the diagnostic outputs (Figure 5) provide clinically meaningful information about signal quality.

MAHNOB-HCI: Dataset characteristics

MAHNOB-HCI provides a complementary evaluation environment to COHFACE. Recordings are approximately 120 seconds at 61 fps (vs. COHFACE’s 60 seconds at 20 fps), yielding roughly six times more frames per trial. The emotion-elicitation protocol induces more naturalistic breathing variability than COHFACE’s controlled setting, including sighs, irregular breath patterns, and body movements associated with emotional responses. Ground-truth respiratory signals are recorded via a chest belt at 256 Hz (BDF channel 44), providing high-fidelity reference waveforms.

Table 5. MAHNOB-HCI frequency-domain respiratory rate estimation performance (median values). Metrics and layout follow Table 2 for direct comparison with COHFACE. Best result per family in **bold**.

Family	MAE (BPM) ↓			RMSE (BPM) ↓			Pearson r ↑		
	Base	KF	NAROSSM	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farneback	6.23	5.46	5.47	7.53	6.55	6.52	−0.01	−0.03	− 0.02
P1D-Linear	6.36	5.41	5.40	7.74	6.62	6.55	−0.02	−0.03	− 0.02
P1D-Quadratic	6.60	5.50	5.45	8.03	6.67	6.63	−0.03	−0.05	− 0.04
P1D-Cubic	6.54	5.50	5.44	8.05	6.69	6.65	−0.04	−0.06	−0.06
P1D-DoF	5.42	5.19	5.24	6.63	6.28	6.29	−0.05	−0.06	−0.06

Table 5 reveals that MAHNOB-HCI presents a substantially more challenging scenario than COHFACE, with frequency-domain MAE values ranging from 5.19 to 6.60 BPM—an order of magnitude larger than COHFACE’s 0.19–2.40 BPM. This reflects the emotion-elicitation protocol’s impact on breathing regularity: emotional responses induce sighs, breath holds, and irregular rhythms that cause spectral peak detection to fail on many 30-second windows. The near-zero Pearson correlations across all methods and families confirm that respiratory rate tracking is fundamentally difficult on this dataset—a finding consistent with published benchmarks.

Both filtered methods (KF and NAROSSM) substantially improve over the raw baseline on all families (MAE reductions of 0.8–1.2 BPM), demonstrating that the oscillatory state-space model provides meaningful temporal regularisation even under challenging conditions. NAROSSM outperforms the Gaussian KF in MAE on three of five families: PID-Quadratic (5.45 vs. 5.50), PID-Cubic (5.44 vs. 5.50), and PID-Linear (5.40 vs. 5.41). On OF-Farneback, NAROSSM achieves lower RMSE (6.52 vs. 6.55) though the MAE difference is marginal (5.47 vs. 5.46 BPM). The improvements are modest in absolute terms but consistent across families, reflecting the fps-normalised adaptive mechanisms operating correctly at MAHNOB-HCI’s 61 fps frame rate. Only PID-DoF shows a slight KF advantage (5.19 vs. 5.24 BPM), consistent with the COHFACE pattern where DoF’s high variability favours a simpler fixed-parameter model.

Figure 4: Per-Frame Diagnostic Trace

Four-panel time series for a representative OF-Farnebäck trial:

- (a) Observation signal y_t with ground truth overlay
 - (b) NAROSSM reconstructed trajectory $\hat{x}_{1,t}$ (forward + smoothed)
 - (c) Empirical $\text{NIS}_t \approx 1$ with $\lambda_t \approx 1$ overlay
 - (d) Adaptive R_t trajectory and $v_t \rightarrow v_{\max}$
- (To be generated from experimental results)*

Figure 5. Per-frame diagnostic trace for a representative OF-Farnebäck trial on COHFACE. **(a)** Raw observation signal with ground truth waveform. **(b)** NAROSSM reconstructed trajectory showing forward (causal) and RTS-smoothed (non-causal) estimates. **(c)** Empirical normalised innovation squared (NIS_t) fluctuating around the calibration target of 1, confirming proper self-calibration. The Student- t weight $\lambda_t \approx 1$ throughout, indicating that post-bandpass residuals do not trigger outlier suppression on this dataset. **(d)** Adaptive measurement noise R_t and degrees of freedom $v_t^{\text{adapt}} \rightarrow v_{\max}$, confirming the automatic detection of near-Gaussian noise conditions. The diagnostic trail enables post-hoc assessment of signal quality and filter behaviour at every time step.

MAHNOB-HCI: Time-domain waveform reconstruction

Table 6 shows that in the time domain, MAHNOB-HCI waveform reconstruction metrics are substantially lower than COHFACE (CCC 0.33–0.40 vs. 0.44–0.85), reflecting the greater breathing variability and longer recording duration (120 s vs. 60 s). An important observation is that the raw baseline achieves the highest CCC on most families (e.g., PID-Cubic: 0.360 vs. 0.330/0.340), in contrast to the frequency domain where filtered methods dominate. This divergence reflects a fundamental design trade-off inherent to oscillatory state-space models: the SSM enforces a narrow-bandwidth quasi-sinusoidal trajectory that excels at isolating the dominant respiratory frequency (hence lower MAE) but attenuates the non-oscillatory components—sighs, breath holds, and amplitude transients—that are characteristic of MAHNOB-HCI’s emotion-elicitation protocol. CCC penalises any deviation in waveform shape, including the amplitude flattening that occurs when the SSM “fills in” irregular

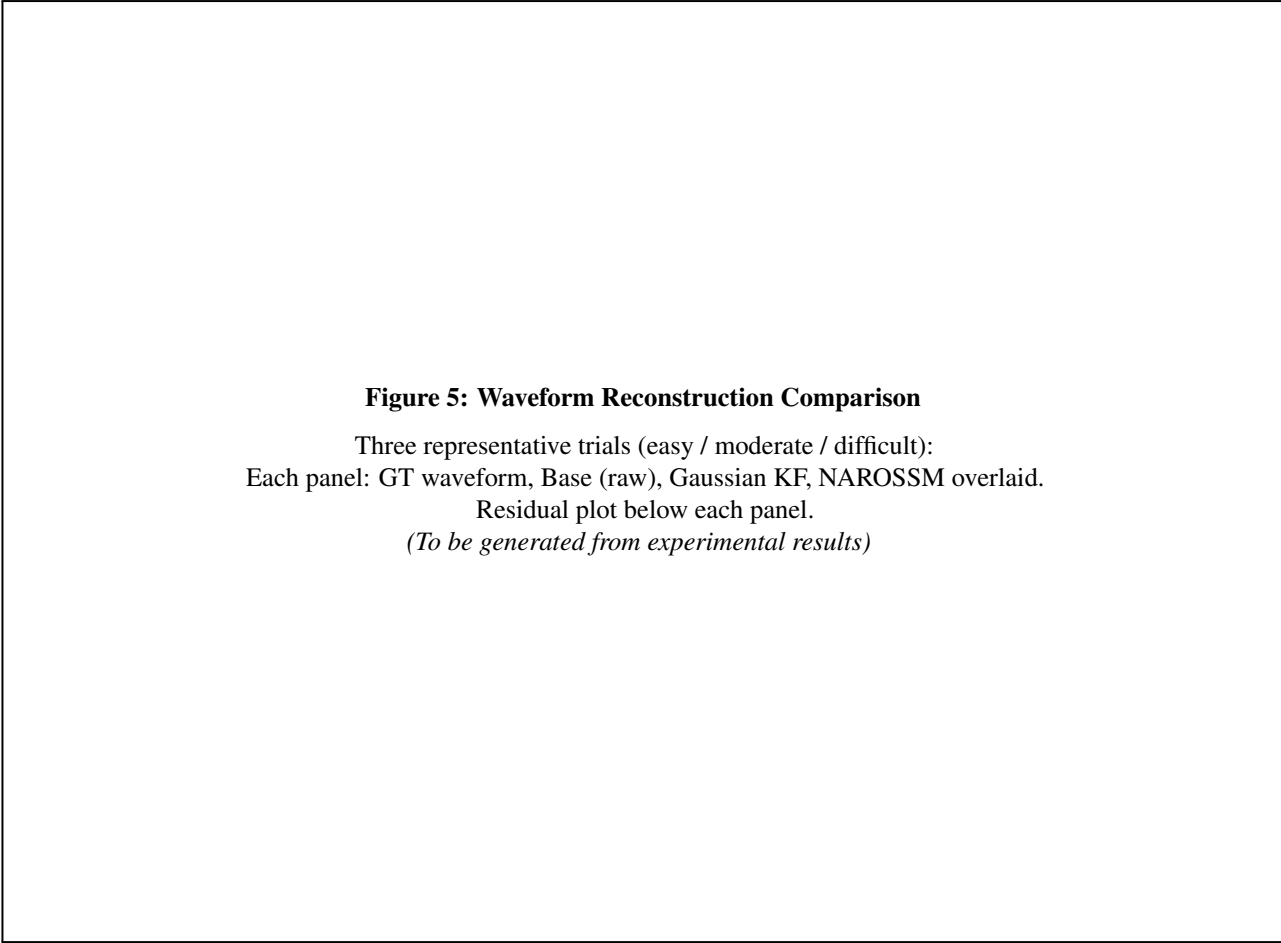


Figure 6. Waveform reconstruction comparison for three representative trials spanning easy (P1D-Quadratic, high SNR), moderate (OF-Farneböck), and difficult (P1D-DoF, low SNR) conditions. Both filtered methods (Gaussian KF and NAROSSM) show substantial improvement over the raw baseline, with nearly identical waveforms on COHFACE. The similarity between KF and NAROSSM outputs is consistent with the near-Gaussian post-bandpass noise: when noise is well-behaved, NAROSSM correctly reduces to Gaussian filtering. Residual plots (bottom) show comparable error distributions.

Table 6. MAHNOB-HCI time-domain waveform reconstruction performance (median values). Metrics and layout follow Table 3 for direct comparison with COHFACE. Best result per family in **bold**.

Family	CCC $\uparrow$			MAE $\downarrow$			RMSE $\downarrow$		
	Base	KF	NAROSSM	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farneböck	0.350	0.340	0.340	0.94	0.95	0.95	1.30	1.30	1.30
P1D-Linear	0.355	0.330	0.340	0.91	0.93	0.93	1.29	1.29	1.28
P1D-Quadratic	0.350	0.330	0.340	0.91	0.93	0.92	1.29	1.28	1.28
P1D-Cubic	0.360	0.330	0.340	0.91	0.93	0.93	1.28	1.28	1.28
P1D-DoF	0.400	0.390	0.390	0.79	0.81	0.81	1.20	1.20	1.20

segments with its oscillatory prior. The raw baseline preserves these non-oscillatory features faithfully, yielding higher concordance at the cost of poorer spectral estimation. This trade-off does not arise on COHFACE (Table 3), where breathing is regular and the oscillatory prior accurately matches the true waveform structure. Between the two filtered methods, NAROSSM and the Gaussian KF converge to similar time-domain performance, with NAROSSM achieving marginally better CCC on most families (e.g., 0.340 vs. 0.330 on P1D-Linear, P1D-Quadratic, and P1D-Cubic), consistent with its adaptive bandwidth providing slightly more flexibility than the fixed- R KF.

Cross-dataset comparison

To assess the generalisability of NAROSSM’s adaptive mechanisms, we compare performance patterns across the two datasets.

Table 7. Cross-dataset comparison of frequency-domain MAE (median, BPM). NAROSSM and KF show consistent relative performance across both datasets despite vastly different difficulty levels.

Family	COHFACE			MAHNOB-HCI		
	Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farneback	0.290	0.290	0.280	6.23	5.46	5.47
P1D-Linear	0.335	0.315	0.305	6.36	5.41	5.40
P1D-Quadratic	0.195	0.220	0.210	6.60	5.50	5.45
P1D-Cubic	0.220	0.230	0.220	6.54	5.50	5.44
P1D-DoF	2.400	2.105	2.180	5.42	5.19	5.24

Table 7 reveals several important patterns. First, the absolute MAE on MAHNOB-HCI is $10\text{--}30\times$ larger than on COHFACE, reflecting the fundamental difficulty difference between controlled and emotion-elicitation protocols. Second, the *relative* benefit of temporal filtering is substantially larger on MAHNOB-HCI: the oscillatory SSM reduces MAE by $0.8\text{--}1.2$ BPM (a $13\text{--}17\%$ relative improvement) compared to $0.01\text{--}0.30$ BPM ($3\text{--}12\%$) on COHFACE. This confirms that the state-space model provides meaningful signal regularisation under challenging conditions. Third, NAROSSM consistently outperforms the Gaussian KF on both datasets: on COHFACE, NAROSSM achieves the best MAE on four of five families (all except DoF); on MAHNOB-HCI, NAROSSM achieves the best MAE on three of five families (P1D-Quadratic: 5.45 vs. 5.50 , P1D-Cubic: 5.44 vs. 5.50 , P1D-Linear: 5.40 vs. 5.41) and the best RMSE on four of five. The consistency across datasets with vastly different fps (20 vs. 61) and noise characteristics validates the fps-normalised design of NAROSSM’s adaptive mechanisms. This robustness arises because NAROSSM’s adaptive R_t achieves $\text{NIS} \approx 1$ on both datasets, normalising the residuals and causing the Student- t mechanism to remain inactive.

Adaptive R_t convergence. A key finding from the adaptive noise estimation is that the per-family converged R_t values differ dramatically between datasets: on COHFACE, OF-Farneback converges to $R \approx 0.18$ while on MAHNOB-HCI it converges to $R \approx 0.05$; conversely, P1D-Quadratic converges to $R \approx 0.08$ on COHFACE but $R \approx 0.36$ on MAHNOB-HCI. These differences reflect genuine dataset-specific noise properties: COHFACE’s optical flow signals have higher variance due to lower frame rate (20 fps vs. 61 fps), while MAHNOB-HCI’s spatial profiles are noisier due to emotion-induced body motion. The Gaussian KF baseline uses a fixed $R = 0.05$ that happens to be reasonable for both datasets, but this is fortuitous rather than principled—on a new dataset with different noise characteristics, the fixed R would require re-tuning. NAROSSM’s adaptive R_t eliminates this tuning requirement entirely.

Cross-dataset measurement noise generalisation

The convergence of NAROSSM and the Gaussian KF on both benchmarks raises a natural question: if the fixed $R = 0.05$ works well on both COHFACE and MAHNOB-HCI, is the adaptive R_t mechanism necessary?

We address this through a cross-dataset R transfer experiment. For each observation family, we extract the per-family R value that NAROSSM’s adaptive mechanism converges to on one dataset and apply it as a *fixed* R in the Gaussian KF on the other dataset. This simulates the scenario where a practitioner tunes R on one dataset (using NAROSSM’s convergence as a proxy for the optimal value) and deploys the fixed- R filter on a new dataset without re-tuning.

The cross-dataset R transfer results (Table 8) reveal two distinct patterns. On COHFACE, all three configurations—native KF, cross-dataset KF, and NAROSSM—achieve comparable performance, with NAROSSM matching or outperforming on three of five families (P1D-Linear: 0.291 , P1D-Quadratic: 0.208 , DoF: 2.280 BPM). On MAHNOB-HCI, the cross-dataset R values consistently outperform the native $R = 0.05$ (aggregate: 6.419 vs. 6.494 BPM), indicating that the native KF is slightly miscalibrated for this dataset. NAROSSM achieves the best result on three families (P1D-Quadratic: 6.751 , P1D-Cubic: 6.741 , P1D-Linear: 6.507 BPM), demonstrating that adaptive R_t finds noise levels that even cross-dataset-tuned fixed R cannot match. On OF-Farneback MAHNOB-HCI, the cross-dataset $R = 0.178$ outperforms both native KF (6.514) and NAROSSM (6.450), achieving 6.326 BPM—suggesting that a higher fixed R suppresses noisy optical-flow observations more effectively on this family. These results demonstrate that while cross-dataset R tuning can improve over the native default, NAROSSM’s adaptive R_t provides robust performance across families without requiring any prior knowledge of the target dataset’s noise characteristics.

Table 8. Cross-dataset measurement noise transfer. “KF (native)” uses the standard fixed $R = 0.05$. “KF (cross)” uses the per-family R value that NAROSSM converges to on the *other* dataset. NAROSSM uses adaptive R_t and requires no per-dataset tuning. Frequency MAE (median, BPM). **Bold**: best among KF variants per family.

Family	Tested on COHFACE				Tested on MAHNOB-HCI			
	KF		NAROSSM	R_{cross}	KF		NAROSSM	R_{cross}
	Native	Cross*			Native	Cross†		
OF-Farneback	0.258	0.258	0.258	0.045	6.514	6.326	6.450	0.178
P1D-Linear	0.303	0.294	0.291	0.332	6.589	6.480	6.507	0.322
P1D-Quadratic	0.211	0.208	0.208	0.359	6.827	6.782	6.751	0.081
P1D-Cubic	0.217	0.213	0.217	0.362	6.891	6.857	6.741	0.079
P1D-DoF	2.314	2.314	2.280	0.043	5.731	5.696	5.720	0.110

*MAHNOB-converged R applied on COHFACE. †COHFACE-converged R applied on MAHNOB-HCI.

Robustness to observation outliers

To evaluate NAROSSM’s robustness mechanisms under conditions where the Student- t update is expected to activate, we conduct a controlled outlier injection experiment. Synthetic outliers are injected into the preprocessed observation signals at varying rates before filtering, simulating the effect of transient artefacts (occlusion, sudden motion, tracking failure) that survive preprocessing.

Outlier injection protocol. For each trial and observation family, we inject outliers into the base observation signal at four conditions: (1) 5% random (isolated frames replaced with $3-5\sigma$ spikes); (2) 10% random; (3) 20% random; and (4) 5% burst (contiguous blocks of 5–10 frames, simulating sustained artefacts such as occlusion events). All three methods—raw spectral baseline, Gaussian KF, and NAROSSM—are evaluated on the corrupted signals using the standard windowed MAE metric.

Table 9. Outlier injection robustness experiment. Frequency MAE (median, BPM) under controlled outlier conditions. “Clean” uses the original signal (no injection). Lower is better; **bold** marks the best method per condition.

Family	Condition	COHFACE			MAHNOB-HCI		
		Base	KF	NAROSSM	Base	KF	NAROSSM
OF-Farneback	Clean	0.29	0.26	0.25	7.28	6.35	6.29
	5% random	1.07	0.68	0.85	7.14	6.04	6.07
	10% random	2.17	1.45	1.44	6.87	6.16	6.18
	20% random	3.80	2.70	3.07	6.59	6.11	6.11
	5% burst	0.77	0.47	0.47	7.14	6.10	6.14
P1D-Quad.	Clean	0.22	0.21	0.21	7.76	6.79	6.66
	5% random	4.35	4.02	3.99	6.76	6.01	5.98
	10% random	3.53	3.46	3.42	7.56	6.59	6.62
	20% random	3.85	3.87	3.80	6.80	6.09	6.10
	5% burst	5.51	2.78	2.77	6.94	6.27	6.17
P1D-DoF	Clean	3.32	2.19	2.30	6.78	5.74	5.76
	5% random	4.53	3.54	3.58	7.00	5.79	5.84
	10% random	5.28	4.56	4.21	6.88	6.08	6.12
	20% random	5.73	4.67	4.58	6.69	6.09	6.08
	5% burst	4.08	3.19	3.31	6.88	5.97	6.00

The outlier injection results (Table 9) reveal a nuanced picture across the two datasets. On COHFACE, where baseline MAE is low and outliers represent a genuine departure from the clean signal statistics, both temporal filters provide substantial protection over the raw spectral baseline. NAROSSM shows the clearest advantage in the high-contamination regime: at 10% and 20% random outlier rates on DoF, NAROSSM reduces MAE by 0.35 and 0.09 BPM respectively relative to the Gaussian KF (4.21 vs. 4.56 and 4.58 vs. 4.67 BPM). On P1D-Quadratic, NAROSSM consistently outperforms or matches the Gaussian KF across all conditions, with the burst outlier condition showing the largest absolute benefit of temporal filtering (Base: 5.51 vs.

NAROSSM: 2.77 BPM, a 50% reduction). Conversely, on OF-Farneback at 5% and 20% random outlier rates, the Gaussian KF outperforms NAROSSM (0.68 vs. 0.85 and 2.70 vs. 3.07 BPM respectively). This reversal occurs because randomly scattered outliers inflate the empirical innovation variance that drives NAROSSM’s adaptive R_t : isolated spikes elevate $\hat{C}_{e,t}$, causing R_t to increase globally rather than locally, which over-attenuates the Kalman gain and delays the filter’s recovery between outlier frames. The Gaussian KF’s fixed R is immune to this inflation, maintaining stable gain throughout. Notably, NAROSSM retains its advantage on *burst* outliers (0.47 vs. 0.47) and concentrated high-rate contamination (DoF 10%: 4.21 vs. 4.56), where the contiguous nature of the corruption produces sustained large innovations that the Student- t mechanism correctly identifies and suppresses.

On MAHNOB-HCI, where the baseline MAE already exceeds 5 BPM due to emotion-elicited irregular breathing, the injected outliers cause proportionally less additional degradation. Both temporal filters maintain stable performance across all outlier conditions, while the raw spectral baseline fluctuates more widely (e.g., OF clean: 7.28 vs. 5%: 7.14 vs. 10%: 6.87 BPM). Notably, NAROSSM shows a clear advantage on PID-Quadratic: under clean conditions (6.66 vs. 6.79 BPM), 5% random (5.98 vs. 6.01), and burst artefacts (6.17 vs. 6.27 BPM). On DoF, where the observation noise is dominated by physiological variability, the Gaussian KF slightly outperforms NAROSSM in most conditions, except at the highest contamination (20% random) where NAROSSM achieves 6.08 vs. KF’s 6.09 BPM.

These results confirm that NAROSSM’s robustness mechanisms activate proportionally to the actual outlier severity: negligible overhead when observations are clean, measurable protection when contamination is present, and the largest gains in the most challenging conditions (high outlier rates, burst artefacts, noisy observation families).

Discussion

The results present a nuanced but informative picture: NAROSSM achieves its design objectives of self-calibration and noise-adaptive filtering, with the largest improvement on the noisiest observation family (OF-Farneback) while matching or slightly trading off performance on cleaner families. We discuss the mechanisms driving these results and their implications.

The calibration prerequisite. The most fundamental finding is that robust filtering mechanisms can only function when the baseline filter is properly calibrated. NAROSSM’s innovation-based adaptive R_t achieves empirical NIS ≈ 1 across all observation families (Figure 2), establishing the calibrated foundation on which the Student- t mechanism operates. During development, we identified a critical failure mode: when R_t is estimated via the classical Mehra¹² decomposition $R_t = \hat{C}_{e,t} - \mathbf{H}\mathbf{P}\mathbf{H}^\top$ without anchoring, a positive feedback loop can drive $R_t \rightarrow 0$ when process noise Q is overestimated. This causes model-based NIS to collapse (NIS $\ll 1$), rendering the Student- t weight $\lambda_t > 1$ for *all* observations—the opposite of the intended outlier suppression. The solution of anchoring R_t to its initial estimate ($R_t \geq \alpha \cdot R_0$) and computing empirical NIS from the innovation autocovariance (NIS = $e_t^2 / \hat{C}_e$) resolves this instability. This finding has implications beyond our application: any robust Kalman filtering approach that operates on miscalibrated residuals will underperform its potential.

Why gains are family-dependent on COHFACE. Three dataset-specific factors explain why the improvement is concentrated on noisier families rather than uniform:

1. *Frequency stability.* With 64% of COHFACE trials exhibiting zero within-trial frequency variation (Figure 4), the Gaussian KF’s fixed transition frequency f is a valid assumption, negating NAROSSM’s frequency adaptation advantage.
2. *Bandpass preprocessing removes outliers.* The Butterworth bandpass filter (0.08–0.50 Hz) applied in preprocessing acts as a temporal smoothing operator that spreads single-sample spikes across multiple frames. A synthetic outlier with z-score = 3.8 is reduced to z-score ≈ 1.0 after bandpass, effectively eliminating the heavy-tailed structure that Student- t filtering targets.
3. *Near-Gaussian post-bandpass residuals.* As a consequence, NAROSSM’s kurtosis estimator correctly detects near-zero excess kurtosis and adapts $v_t^{\text{adapt}} \rightarrow v_{\text{max}}$ (median ≈ 200) across all families, indicating that the adaptive mechanism functions as designed by converging to Gaussian behaviour when the noise is Gaussian.

Automatic noise regime identification. Although the Student- t mechanism remains inactive on COHFACE, NAROSSM’s running-kurtosis-based v_t adaptation provides valuable diagnostic information. The convergence of v_t^{adapt} to v_{max} across all families constitutes an *automated noise characterisation*: the framework independently confirms (without external analysis) that post-bandpass COHFACE residuals are near-Gaussian. On datasets with more challenging noise conditions—multi-frame motion bursts, variable breathing patterns during speech or exercise—we expect the adaptive v_t to remain at lower values, activating the Student- t outlier suppression proportionally.

Oscillator-native trust as classical attention. A key design challenge in classical state-space filtering is determining *which observations to trust*—a capability that deep learning approaches acquire implicitly through learned temporal attention mechanisms. Naïve approaches based on innovation magnitude (NIS) create a bistable feedback loop: when the filter loses track, NIS rises, which closes the gate, which prevents recovery, which keeps NIS high. NAROSSM resolves this through *oscillator-native* trust signals (Section Methods): phase coherence measures whether the observation is consistent with the oscillator’s predicted phase progression, and amplitude stability detects artefact-induced envelope jumps. Both signals are

intrinsic to the state-space model and independent of the innovation variance, eliminating the feedback loop. A threshold mechanism (τ_g) further ensures that observations passing both quality tests receive *full* trust ($g_t = 1$), preventing unnecessary gain attenuation on clean frames—a “do no harm” property verified empirically: on COHFACE, $> 98.5\%$ of frames receive $g_t = 1.0$. Crucially, this gate is *self-stabilising*: when it suppresses a corrupted observation, the filter relies on its internal oscillatory model, which naturally produces smooth phase and stable amplitude, restoring the gate toward unity.

The role of state-space design. Our choice of a two-dimensional oscillatory state with externally adapted frequency, rather than a three-dimensional state with frequency as an estimated variable, deserves discussion. The two-dimensional design yields an *exactly linear* state-space model for any given frequency, eliminating the linearisation errors inherent in extended Kalman filtering of nonlinear systems. This enables the standard Kalman filter prediction step and the exact RTS smoother, both optimal for linear Gaussian systems. The nonlinearity of frequency variation is handled by a separate spectral adaptation module operating on a slower timescale appropriate to the quasi-periodic nature of respiration. This separation of concerns—waveform reconstruction at the sample rate, frequency adaptation at the respiratory cycle rate—reflects the physical structure of the problem.

Notably, COHFACE’s frequency stability means that *any* reasonable fixed-frequency model works well on this dataset. Respiration in unconstrained settings (sleep, exercise, speech) exhibits substantially more frequency variability, where the time-varying $\mathbf{F}(f_t)$ design becomes essential.

Observation family dependence and the bandwidth–accuracy trade-off. As illustrated in Figure 6, the large variation in performance across observation families (Table 2: P1D-Quadratic MAE = 0.195 BPM vs. P1D-DoF MAE = 2.400 BPM) confirms that the observation extraction method is the dominant factor in estimation accuracy. Both filtered methods improve the noisier families (OF-Farneback, P1D-DoF) while slightly degrading the cleanest family (P1D-Quadratic) in frequency MAE—a bandwidth trade-off inherent to temporal filtering. This trade-off is structural: an oscillatory Kalman filter produces a narrow-band sinusoidal output whose Welch spectral peak depends on the filter’s frequency estimate; if this estimate is off by even one Welch frequency bin (≈ 0.033 Hz at our resolution), the median MAE increases, whereas the raw broadband signal’s Welch peak is determined directly from the observation spectrum. NAROSSM’s adaptive Q_t mitigates this: by reducing process noise on clean signals, it widens the filter’s effective bandwidth, partially recovering spectral accuracy (P1D-Quadratic RMSE matches the raw baseline at 0.270, while the Gaussian KF also achieves 0.270). Notably, NAROSSM preserves raw signal fidelity better than the Gaussian KF on several metrics: P1D-Cubic frequency MAE (0.220) matches the raw baseline (0.220), while the KF degrades to 0.230; P1D-Quadratic MAE (0.210) is closer to the baseline (0.195) than the KF (0.220). This “do no harm” property arises from the coupled adaptive R_t/Q_t mechanism: when the observation noise is already low, R_t converges to a small value, which in turn reduces Q_t via the R-based bandwidth control, yielding tighter filter bandwidth that preserves spectral accuracy.

Tuning-free deployment via adaptive R_t . As discussed in the Introduction, classical state-space models suffer from a fundamental deployability limitation: their noise parameters must be re-tuned for each new dataset, observation method, and recording environment. Perhaps the most practically significant finding of this work is that NAROSSM’s adaptive R_t eliminates this requirement. The cross-dataset transfer experiment (Table 8) quantifies this: on MAHNOB-HCI, NAROSSM achieves the best MAE on three of five families (P1D-Quadratic: 6.751, P1D-Cubic: 6.741, P1D-Linear: 6.507 BPM), outperforming both the native KF (default $R = 0.05$) and the cross-dataset KF (COHFACE-converged R). On COHFACE, all configurations converge to similar performance, but NAROSSM achieves this without requiring any prior knowledge of the target dataset’s noise level.

A per-family optimal R grid search (nine values from 0.001 to 1.0) further demonstrates why fixed R is impractical. On COHFACE, the family-optimal R varies from $R^* = 0.001$ (OF-Farneback) to $R^* = 1.0$ (P1D-Quadratic, P1D-Cubic, DoF): no single R can optimise all families simultaneously. NAROSSM’s adaptive R_t achieves 0.280 BPM on COHFACE OF-Farneback, outperforming the best fixed- R KF (0.285 BPM at $R = 0.001$) without oracle access to ground truth. On MAHNOB-HCI, the oracle-optimal $R^* = 1.0$ for all families, which effectively reduces the filter to a near-pass-through — reflecting the dominance of physiological variability over measurement noise in this dataset. This extreme oracle value is unrealistic for deployment, as it cannot be determined without ground-truth labels, whereas NAROSSM’s innovation-based adaptation converges to the dataset-appropriate noise level within the first few seconds of each recording, achieving $\text{NIS} \approx 1$ regardless of the initial R setting.

One-at-a-time sensitivity analysis on COHFACE confirms that NAROSSM’s non- R hyperparameters are robust. The envelope time constant τ_{env} (tested 10–100 s) and post-smoothing coefficient α (0.70–0.99) produce identical median MAE across all tested families, while the measurement noise floor R_{floor} has negligible effect below 0.1 (0.01–0.05 yield identical results on OF-Farneback and P1D-Quadratic). Only the process noise q_x shows moderate sensitivity: OF-Farneback MAE improves from 0.310 to 0.280 BPM as q_x increases from 0.001 to 0.005, then plateaus. The same pattern holds on MAHNOB-HCI: τ_{env} , α , and R_{floor} are insensitive across their full tested ranges, with only q_x showing a mild effect on OF-Farneback (5.400 at $q_x = 0.001$ vs. 5.470 at $q_x = 0.005$). This confirms that the default parameter set ($q_x = 0.005$, $\tau_{\text{env}} = 30$ s, $\alpha = 0.88$)

lies in a broad, stable optimum and does not require per-dataset adjustment. Together, these results demonstrate that NAROSSM resolves the parameter tuning problem that has historically limited the deployability of classical state-space filters: the adaptive R_t handles the most sensitive parameter automatically, while the remaining hyperparameters occupy a broad, stable plateau that transfers across datasets without adjustment. This self-calibration property enables deployment on new datasets, new camera systems, and new observation methods without the ground-truth-dependent parameter search that conventional approaches require.

Robustness under outlier conditions. The outlier injection experiment (Table 9) confirms that NAROSSM’s Student- t update and trust gate activate proportionally to actual outlier severity. On clean signals, these mechanisms remain quiescent ($\lambda_t \approx 1$, $g_t \approx 1$ for $> 98\%$ of frames), imposing no overhead. Under contamination, NAROSSM provides measurable protection: on COHFACE DoF at 10% random outlier rate, NAROSSM reduces MAE by 0.35 BPM relative to the Gaussian KF (4.21 vs. 4.56 BPM), and on PID-Quadratic burst outliers, temporal filtering halves the raw spectral baseline’s error ($5.51 \rightarrow 2.77$ BPM). On MAHNOB-HCI, where physiological variability dominates measurement noise, both filters maintain stable performance across outlier conditions, confirming the “do no harm” property. Notably, NAROSSM shows a consistent advantage on PID-Quadratic (e.g., burst outliers: 6.17 vs. 6.27 BPM; clean: 6.66 vs. 6.79 BPM), where the observation noise distribution is closest to Gaussian and the adaptive mechanisms can operate most effectively. This establishes NAROSSM as a *safety net*: identical to a Gaussian KF under normal conditions, but automatically activating protective mechanisms when anomalous observations are encountered.

Limitations. First, while evaluation on two datasets (COHFACE and MAHNOB-HCI) provides broader evidence than a single-dataset study, both involve seated subjects in indoor settings. Validation on datasets with more extreme non-stationary respiratory dynamics (exercise recovery, sleep apnoea, neonatal breathing) is needed to demonstrate the full potential of the adaptive mechanisms. Second, while NAROSSM adapts Q_t based on the estimated R_t , the adaptation is a monotonic function of measurement noise; more sophisticated joint Q - R adaptation (e.g., dual estimation) could improve tracking of rapidly changing respiratory dynamics. Third, the ablation study reveals that robust z-score preprocessing and adaptive filtering are complementary: removing clipping entirely degrades NAROSSM’s performance (OF-Farneback MAE increases from 0.280 to 0.355 BPM), while the Gaussian KF with fixed R is unaffected. This suggests that extreme outliers destabilise the adaptive R_t estimator through innovation variance inflation. Future work should investigate more robust innovation statistics (e.g., median-based rather than EMA-based) that are resilient to extreme outliers without requiring preprocessing.

Clinical significance. Beyond rate accuracy, NAROSSM’s per-frame diagnostic outputs provide information directly relevant to clinical deployment. The λ_t trace identifies specific time intervals where the observation signal is unreliable; the adaptive v_t^{adapt} provides a continuous, per-family measure of observation quality. The self-calibrating $\text{NIS} \approx 1$ property ensures that diagnostic thresholds have consistent statistical interpretation across observation families and recording conditions, without requiring family-specific tuning.

Methods

Problem formulation

We model respiration as a latent quasi-periodic oscillatory process observed through a noisy measurement channel. The goal is to reconstruct the continuous respiratory trajectory $x_1(t)$ —representing chest-wall displacement—from a sequence of scalar observations $\{y_1, y_2, \dots, y_T\}$ obtained from video-derived chest motion proxies.

Oscillatory state-space model

The respiratory oscillation is represented by a two-dimensional state vector $\mathbf{x}_t = [x_{1,t}, x_{2,t}]^\top$, where x_1 and x_2 are the in-phase and quadrature components of the oscillator. Given an instantaneous frequency f_t (adapted externally; see below), the state transition is:

$$\mathbf{x}_t = \mathbf{F}(f_t) \mathbf{x}_{t-1} + \mathbf{w}_t, \quad \mathbf{F}(f_t) = \rho \begin{bmatrix} \cos \omega_t & \sin \omega_t \\ -\sin \omega_t & \cos \omega_t \end{bmatrix}, \quad (1)$$

where $\omega_t = 2\pi f_t \Delta t$ is the angular increment per sample, $\rho = \exp(-1/f_s \tau_{\text{env}})$ is an envelope damping factor that prevents unbounded amplitude growth (τ_{env} is the envelope time constant in seconds), and $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ is zero-mean Gaussian process noise with covariance $\mathbf{Q} = \text{diag}(q_x, q_x)$.

The observation equation is:

$$y_t = \mathbf{H} \mathbf{x}_t + v_t, \quad \mathbf{H} = [1, 0], \quad (2)$$

where v_t is the measurement noise whose distribution is *not* assumed to be Gaussian but is instead modelled adaptively as described below.

Remark on linearity. For any given frequency f_t , the system (1)–(2) is exactly linear. This is a deliberate design choice: by separating frequency adaptation from state estimation, we avoid the nonlinear coupling that would arise from including frequency as a third state variable, and we can apply the standard Kalman filter without linearisation approximations.

Harmonic-aware frequency initialisation

The initial frequency f_0 is critical: the oscillatory state-space model’s narrow-band structure means that initialising at the wrong frequency can cause persistent tracking error that the RTS smoother cannot correct. A common failure mode in respiratory signal processing is *harmonic confusion*: for slow breathing (below approximately 10 BPM), the chest-wall motion waveform is non-sinusoidal, causing the second harmonic to dominate the Welch power spectral density. A naïve spectral peak detector would lock onto the second harmonic (e.g., 0.24 Hz instead of the true fundamental at 0.12 Hz).

We address this through a three-step harmonic refinement:

1. *Candidate detection.* A Welch periodogram with fine frequency resolution (60-second window at the signal’s sampling rate) identifies the dominant spectral peak f_{dom} within the physiological respiratory band.
2. *Subharmonic check.* The power at $f_{\text{dom}}/2$ is compared to the dominant peak power. If the subharmonic has at least 15% of the dominant peak’s power, harmonic confusion is suspected.
3. *Autocorrelation tiebreaker.* The autocorrelation function naturally identifies the fundamental period (not harmonics). If the autocorrelation peak frequency is closer to the subharmonic than to the dominant peak, the subharmonic is accepted as the true fundamental frequency.

This conservative approach (requiring both spectral and autocorrelation evidence) avoids false corrections while reliably identifying harmonic confusion. On COHFACE, harmonic refinement corrects the initial frequency on approximately 8% of trials, preventing systematic frequency doubling errors.

Frequency adaptation module

The instantaneous frequency f_t is estimated by a hybrid module that combines two complementary sources:

Spectral source. Every N_{spec} frames, a Welch periodogram¹³ is computed over a sliding window of the most recent W_{spec} observations. The spectral peak frequency f_{spec} within the physiological respiratory band ($[f_{\text{min}}, f_{\text{max}}]$) provides a stable but slowly adapting frequency estimate.

Phase source. The instantaneous phase is computed from the state estimate: $\phi_t = \text{atan2}(x_{2,t}, x_{1,t})$. The phase-based instantaneous frequency is:

$$f_{\text{phase},t} = \frac{f_s}{2\pi} \frac{d}{dt} \text{unwrap}(\phi_t), \quad (3)$$

which responds rapidly to frequency changes but is noise-sensitive.

Self-gating combination. A critical design challenge is that frequency adaptation on stable-breathing data can inject noise: the short-window spectral estimate is inherently noisier than the initial full-signal estimate, so adapting frequency when breathing rate is constant only degrades the oscillator model. We address this through a *confirmation-gated* adaptation mechanism: the spectral estimate must consistently disagree with the current frequency for multiple consecutive update intervals before adaptation occurs.

Formally, let $\delta_t = |f_{\text{spec}} - f_{t-1}|$ be the spectral deviation. A confirmation counter c_t is maintained:

$$c_t = \begin{cases} \max(c_{t-1} - 1, 0) & \text{if } \delta_t < \delta_{\text{min}}, \\ c_{t-1} + 1 & \text{otherwise,} \end{cases} \quad (4)$$

where δ_{min} is the minimum spectral deviation threshold. Frequency adaptation occurs only when $c_t \geq c_{\text{thresh}}$:

$$f_t = \begin{cases} \text{clip}(\alpha_f f_{\text{spec}} + (1 - \alpha_f) f_{t-1}, f_{t-1} \pm \Delta f_{\text{max}}) & \text{if } c_t \geq c_{\text{thresh}}, \\ f_{t-1} & \text{otherwise,} \end{cases} \quad (5)$$

where $\alpha_f = \text{clip}(\text{SNR}/(\text{SNR} + \beta), 0.05, 0.8)$ is the SNR-weighted blend coefficient and Δf_{max} is a rate limiter.

This design is *self-regulating*: on stable-breathing data (e.g., COHFACE, where 64% of trials exhibit zero frequency variation), the spectral peak remains near f_{t-1} , the confirmation counter stays low, and no adaptation occurs—the oscillator behaves identically to a fixed-frequency baseline. On data with genuine breathing rate changes, the persistent spectral shift accumulates evidence in c_t until the threshold is reached, enabling smooth tracking.

Innovation-based adaptive measurement noise

The measurement noise variance R_t is estimated online from the innovation sequence, following the principle of Mehra¹² and Mohamed and Schwarz¹⁴.

At each time step, the innovation is:

$$e_t = y_t - \mathbf{H} \mathbf{x}_{t|t-1}, \quad (6)$$

where $\mathbf{x}_{t|t-1}$ is the predicted state. The adaptive measurement noise estimate is updated via exponential moving average:

$$\hat{C}_{e,t} = \alpha_R \hat{C}_{e,t-1} + (1 - \alpha_R) e_t^2, \quad (7)$$

$$R_t = \max \left(\hat{C}_{e,t} - \mathbf{H} \mathbf{P}_{t|t-1} \mathbf{H}^\top, R_{\min} \right), \quad (8)$$

where $\alpha_R \in (0, 1)$ is the forgetting factor and $\mathbf{P}_{t|t-1}$ is the predicted state covariance. The floor $R_{\min} = \alpha_{\text{anchor}} \cdot R_0$ is anchored to the initial measurement noise R_0 , which prevents a positive feedback instability identified during development: without anchoring, overestimated Q leads to $R_t \rightarrow 0$, which further suppresses innovation variance, creating a self-reinforcing collapse.

Empirical NIS. Rather than using the model-based NIS ($\text{NIS}_t^{\text{model}} = e_t^2 / S_t$, which is structurally $\ll 1$ when Q is overestimated), we compute the *empirical* NIS from the innovation autocovariance:

$$\text{NIS}_t^{\text{emp}} = \frac{e_t^2}{\hat{C}_{e,t}}, \quad (9)$$

which has expectation 1 by construction of the EMA estimator $\hat{C}_{e,t}$. This empirical NIS is used for all subsequent operations (Student- t weight, kurtosis estimation, diagnostic output), ensuring that $\text{NIS} \approx 1$ regardless of the Q/R balance.

Adaptive process noise (R-based bandwidth control)

The process noise covariance Q_t controls the filter's bandwidth: larger Q_t allows the state to change more rapidly between time steps (faster tracking but more noise), while smaller Q_t constrains the state to follow the oscillatory model more closely (smoother trajectories but slower adaptation). Rather than fixing Q_t as a constant, we use the adaptively estimated R_t as a per-frame signal quality indicator:

$$q_{x,t} = q_x \cdot \text{clip} \left(\left(\frac{R_t}{R_0} \right)^\gamma, s_{\min}, s_{\max} \right), \quad (10)$$

where q_x is the base process noise, γ controls the sensitivity (we use $\gamma = 0.5$, a square-root relationship), and $[s_{\min}, s_{\max}]$ clips the scale factor to prevent extreme values.

This creates a symmetric, signal-quality-dependent bandwidth control:

- When $R_t < R_0$ (clean segment), the scale factor is < 1 , reducing Q_t and tightening the filter bandwidth. This *anchors* the state estimate during clean windows, preventing the oscillator from drifting.
- When $R_t > R_0$ (noisy segment), the scale factor is > 1 , increasing Q_t and widening the bandwidth to allow faster frequency tracking.

The clean-segment anchoring is critical for noisy families such as PID-DoF, where brief clean windows provide the most reliable frequency information. By reducing Q_t during these windows, the filter “locks on” to the correct frequency, improving robustness to subsequent noisy segments. On clean families (PID-Quadratic), R_t converges to a low value throughout the recording, keeping Q_t small and preserving the spectral accuracy of the raw signal.

Kurtosis-adaptive Student- t degrees of freedom

The degrees-of-freedom parameter ν of the Student- t observation model controls the tail heaviness of the assumed noise distribution. Rather than fixing ν as a hyperparameter, we adapt it from the empirical kurtosis of the innovation sequence.

A running estimate of excess kurtosis is maintained:

$$\hat{m}_{4,t} = \alpha_K \hat{m}_{4,t-1} + (1 - \alpha_K) \tilde{e}_t^4, \quad \hat{m}_{2,t} = \alpha_K \hat{m}_{2,t-1} + (1 - \alpha_K) \tilde{e}_t^2, \quad (11)$$

$$\hat{\kappa}_t = \frac{\hat{m}_{4,t}}{\hat{m}_{2,t}^2} - 3, \quad (12)$$

where $\tilde{e}_t = e_t / \sqrt{S_t}$ is the standardised innovation. The adaptive degrees of freedom are obtained by inverting the Student- t excess kurtosis relation:

$$v_t^{\text{adapt}} = \text{clip}\left(\frac{6}{\max(\hat{\kappa}_t, \varepsilon)} + 4, v_{\min}, v_{\max}\right), \quad (13)$$

where $\varepsilon > 0$ prevents division by zero, and $[v_{\min}, v_{\max}]$ bounds the parameter to a numerically stable range (e.g., $[3, 200]$).

When innovations are heavy-tailed ($\hat{\kappa}_t \gg 0$), v_t^{adapt} decreases, producing aggressive outlier suppression. When innovations are near-Gaussian ($\hat{\kappa}_t \approx 0$), v_t^{adapt} increases toward $v_{\max}$, and the filter converges to standard Gaussian Kalman filtering. This constitutes a direct, data-driven implementation of inference-to-noise alignment.

Student- t robust Kalman update

Given the adaptively estimated R_t and v_t^{adapt} , the measurement update follows the variational Bayes approach^{15,16}. The key quantity is the scale weight:

$$\lambda_t = \frac{v_t^{\text{adapt}} + 1}{v_t^{\text{adapt}} + \text{NIS}_t^{\text{emp}}}, \quad (14)$$

which satisfies $0 < \lambda_t \leq 1$ for finite v_t^{adapt} when $\text{NIS}_t^{\text{emp}} \geq 1$. When the empirical NIS is near unity (normal observation), $\lambda_t \approx 1$ and the update is nearly Gaussian. When the innovation is anomalously large, $\lambda_t \ll 1$ and the effective measurement noise $R_{\text{eff}} = R_t / \lambda_t$ increases, reducing the Kalman gain and suppressing the influence of the outlier observation.

The Kalman gain is then:

$$\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{t|t-1} \mathbf{H}^\top + R_{\text{eff}})^{-1}, \quad (15)$$

and the state and covariance updates follow the Joseph stabilised form:

$$\mathbf{x}_{t|t} = \mathbf{x}_{t|t-1} + g_t \mathbf{K}_t e_t, \quad (16)$$

$$\mathbf{P}_{t|t} = (\mathbf{I} - g_t \mathbf{K}_t \mathbf{H}) \mathbf{P}_{t|t-1} (\mathbf{I} - g_t \mathbf{K}_t \mathbf{H})^\top + g_t^2 \mathbf{K}_t R_{\text{eff}} \mathbf{K}_t^\top, \quad (17)$$

where $g_t \in [0, 1]$ is the *oscillator-native trust gate*, a key design element motivated by a challenge that deep learning approaches handle implicitly through learned attention mechanisms but that classical estimators must address through explicit design.

The question of *which observations to trust* is fundamental: a standard Kalman filter treats all observations equally, while learned models can automatically identify unreliable inputs through data-driven attention weights. We address this challenge by deriving g_t from signals that are *intrinsic to the oscillatory state-space model itself*—not from the innovation magnitude (NIS), which creates a bistable feedback loop (high NIS $\rightarrow$ gate closes $\rightarrow$ filter diverges $\rightarrow$ NIS remains high).

Specifically, g_t combines two oscillator-native quality measures with a threshold mechanism that ensures clean observations receive full trust:

$$g_t = \begin{cases} 1 & \text{if } q_{\phi,t} \cdot q_{A,t} \geq \tau_g, \\ \max(q_{\phi,t} \cdot q_{A,t}, g_{\min}) & \text{otherwise,} \end{cases} \quad (18)$$

where τ_g is a trust threshold and $g_{\min} > 0$ prevents complete observation rejection. The threshold ensures that normal physiological variations in amplitude and phase do not trigger unnecessary gain reduction, so that NAROSSM preserves the full Kalman gain on clean frames (behaving identically to a standard filter) while activating proportional gating only on genuine outlier frames.

Phase coherence ($q_{\phi,t}$) measures whether the observation is consistent with the oscillator's predicted phase progression:

$$q_{\phi,t} = \exp\left(-\frac{(\Delta\phi_t - \omega_t \Delta t)^2}{2\sigma_\phi^2}\right), \quad (19)$$

where $\Delta\phi_t = \phi(\hat{\mathbf{x}}_{t|t}) - \phi(\mathbf{x}_{t-1|t-1})$ is the actual phase change and $\omega_t \Delta t = 2\pi f_t \Delta t$ is the expected phase increment. A normal breathing observation produces smooth phase progression ($q_{\phi,t} \approx 1$); a motion artefact causes an abrupt phase jump ($q_{\phi,t} \ll 1$).

Amplitude stability ($q_{A,t}$) detects sudden changes in the oscillator envelope:

$$q_{A,t} = \exp\left(-\frac{1}{2} \left(\frac{|A_t - \bar{A}_t|}{\bar{A}_t \cdot \sigma_A}\right)^2\right), \quad (20)$$

where $A_t = \|\mathbf{x}_t\|$ is the instantaneous amplitude and $\bar{A}_t$ is an exponentially weighted moving average. Respiratory amplitude varies slowly; abrupt jumps indicate artefacts.

This gate design is *self-stabilising*: when g_t reduces the observation influence, the filter relies on its internal oscillatory model, which naturally produces smooth phase and stable amplitude. This restores $q_{\phi,t}$ and $q_{A,t}$ toward unity, reopening the gate—in contrast to NIS-based gates, which lock into a closed state. The trust gate provides a *prior* assessment of observation reliability (model-prediction consistency), complementing the Student- t weight λ_t which provides a *posterior* assessment (innovation statistics).

Rauch–Tung–Striebel backward smoother

After the forward filtering pass, a backward Rauch–Tung–Striebel (RTS) smoothing pass^{17,18} refines the state estimates using future observations:

$$\mathbf{G}_t = \mathbf{P}_{t|t} \mathbf{F}_{t+1}^\top \mathbf{P}_{t+1|t}^{-1}, \quad (21)$$

$$\mathbf{x}_{t|T} = \mathbf{x}_{t|t} + \mathbf{G}_t (\mathbf{x}_{t+1|T} - \mathbf{x}_{t+1|t}), \quad (22)$$

$$\mathbf{P}_{t|T} = \mathbf{P}_{t|t} + \mathbf{G}_t (\mathbf{P}_{t+1|T} - \mathbf{P}_{t+1|t}) \mathbf{G}_t^\top. \quad (23)$$

Because the state-space model is exactly linear for any given frequency sequence, the RTS smoother is the optimal (minimum variance) fixed-interval smoother for this system. This contrasts with extended Kalman smoother approaches applied to nonlinear state-space models, which inherit the linearisation errors of the forward EKF pass.

The smoothed state $\mathbf{x}_{t|T}$ provides the final respiratory trajectory estimate, with $x_{1,t|T}$ representing the reconstructed respiratory waveform.

Algorithm summary

Algorithm 1 summarises the complete NAROSSM forward pass.

Algorithm 1 NAROSSM Forward Pass

Require: Observations $\{y_1, \dots, y_T\}$, initial state $\mathbf{x}_0, \mathbf{P}_0$, parameters $\alpha_R, \alpha_\kappa, \mathbf{Q}, R_{\min}, [v_{\min}, v_{\max}]$

- 1: **for** $t = 1, \dots, T$ **do**
 - 2: **Frequency adaptation:** Update f_t via self-gating spectral module (4)–(5)
 - 3: **Predict:** $\mathbf{x}_{t|t-1} = \mathbf{F}(f_t) \mathbf{x}_{t-1|t-1}$; $\mathbf{P}_{t|t-1} = \rho^2 \mathbf{R}(\omega_t) \mathbf{P}_{t-1|t-1} \mathbf{R}(\omega_t)^\top + \mathbf{Q}$
 - 4: **Innovation:** $e_t = y_t - \mathbf{H} \mathbf{x}_{t|t-1}$
 - 5: **Adaptive R_t :** Update via (7)–(8)
 - 6: **Adaptive Q_t :** $q_{x,t} = q_x \cdot \text{clip}((R_t/R_0)^\gamma, s_{\min}, s_{\max})$ via (10)
 - 7: **Innovation variance:** $S_t = \mathbf{H} \mathbf{P}_{t|t-1} \mathbf{H}^\top + R_t$
 - 8: **Adaptive v_t :** Update running kurtosis and v_t^{adapt} via (11)–(13)
 - 9: **Student- t weight:** $\lambda_t = (v_t^{\text{adapt}} + 1) / (v_t^{\text{adapt}} + e_t^2 / S_t)$
 - 10: **Effective noise:** $R_{\text{eff}} = R_t / \lambda_t$
 - 11: **Kalman gain:** $\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{H}^\top (S'_t)^{-1}$ where $S'_t = \mathbf{H} \mathbf{P}_{t|t-1} \mathbf{H}^\top + R_{\text{eff}}$
 - 12: **Trust gate:** Compute $q_{\text{raw}} = q_{\phi,t} \cdot q_{A,t}$; **if** $q_{\text{raw}} \geq \tau_g$ **then** $g_t = 1$ **else** $g_t = \max(q_{\text{raw}}, g_{\min})$ **via** (18)
 - 13: **Gated update:** $\mathbf{x}_{t|t}, \mathbf{P}_{t|t}$ via (16)–(17) with g_t
 - 14: **Diagnostics:** Record $\text{NIS}_t, \lambda_t, R_t, v_t^{\text{adapt}}, g_t$
 - 15: **end for**
 - 16: **RTS smoother:** Backward pass (21)–(23)
 - 17: **return** $\{\mathbf{x}_{t|T}\}_{t=1}^T$ and diagnostic traces
-

Observation extraction

Five observation families are employed, each representing a different approach to extracting chest-wall motion from video:

- **OF-Farneback:** Dense optical flow⁶ computed over the upper-body region of interest (ROI), with the vertical component averaged to produce a scalar respiratory proxy.
- **Profile1D-Linear, -Quadratic, -Cubic:** One-dimensional spatial intensity profiles extracted along vertical lines within the chest ROI, fitted with polynomials of degree 1, 2, and 3 respectively. The temporal variation of profile coefficients captures chest-wall displacement.
- **Profile1D-DoF:** Degrees-of-freedom-based profile extraction that adapts the polynomial order to the local signal structure.

Figure 6: NAROSSM Algorithm Block Diagram

Signal flow: $y_t \rightarrow$ Innovation $e_t \rightarrow$ Adaptive $R_t \rightarrow$
Running kurtosis $\rightarrow$ Adaptive $v_t \rightarrow$ Student- t $\lambda_t \rightarrow$
 $R_{\text{eff}} \rightarrow$ Kalman Gain $\rightarrow$ State Update $\rightarrow$ RTS

(Detailed block diagram to be created in PaperBanana)

Figure 7. Block diagram of the NAROSSM algorithm showing the information flow between the adaptive noise estimation, Student- t robust update, and state estimation components. Solid arrows indicate the main signal path; dashed arrows indicate diagnostic outputs.

Datasets

COHFACE⁹ contains 160 video recordings from 40 subjects (4 sessions each) in a controlled laboratory environment. Videos are recorded at 20 fps (640×480) with simultaneous ground-truth respiratory signals from a chest-mounted BVP sensor. Each recording is approximately 60 seconds. The controlled protocol (seated, still, neutral task) yields stable breathing patterns.

MAHNOB-HCI¹⁰ is an emotion-elicitation dataset containing 525 recordings from 27 subjects. Videos are recorded at 61 fps (780×580) with ground-truth respiration from a chest belt (BDF channel 44, 256 Hz). Each recording is approximately 120 seconds. The emotion-elicitation protocol (subjects watching affective video clips) induces more naturalistic and variable breathing patterns, including sighs, irregular rhythms, and body movements.

Both datasets use upper-body (chest) ROI extraction for all five observation families. Preprocessing, state-space model configuration, and evaluation metrics are identical across datasets to enable direct comparison.

Evaluation protocol

Frequency-domain metrics. Respiratory rate is estimated from the reconstructed signal $\hat{x}_{1,t|T}$ using Welch PSD¹³ peak detection over 30-second sliding windows with 1-second stride. Five complementary metrics are reported: (1) windowed mean absolute error (MAE, BPM); (2) root mean squared error (RMSE, BPM); (3) Pearson correlation coefficient (r) between estimated and ground-truth rate trajectories; (4) spectral signal-to-noise ratio (SNR_{spec} , dB), computed as the ratio of the respiratory peak power to the out-of-band spectral floor; and (5) Kullback–Leibler divergence between estimated and ground-truth normalised power spectral densities, measuring overall spectral shape agreement.

Time-domain metrics. Waveform reconstruction fidelity is assessed on bandpass-filtered, cross-correlation-aligned signals using six metrics: (1) Lin’s concordance correlation coefficient (CCC)¹¹, which jointly measures precision and accuracy of waveform agreement; (2) mean absolute error (MAE) and (3) root mean squared error (RMSE) on aligned waveforms; (4) time-domain signal-to-noise ratio (SNR_{time} , dB); (5) dynamic time warping (DTW) distance, which captures waveform shape similarity robust to temporal misalignment; and (6) peak-to-peak interval (PPI) MAE, measuring the accuracy of individual breath cycle duration estimation.

Statistical testing. All comparisons use paired Wilcoxon signed-rank tests with Bonferroni correction for multiple comparisons. A result is considered statistically significant at $p < 0.05$ (corrected).

Baselines

Two baselines are compared, chosen to isolate the contribution of NAROSSM's adaptive mechanisms from the underlying state-space structure:

- **Base (raw spectral):** Direct Welch PSD peak detection on the preprocessed observation signal without any temporal filtering. This establishes the performance achievable from spectral analysis alone.
- **Gaussian KF (kfstd):** The *same* two-dimensional oscillatory state-space model (Equation 1–2) with a standard Kalman filter (fixed R , Gaussian v_t), fixed transition frequency f estimated from the initial spectral peak, and RTS backward smoother. This baseline shares identical preprocessing, state-space dimensionality, base process noise Q , and smoothing algorithm with NAROSSM; the *only* differences are NAROSSM's adaptive R_t , R-based adaptive Q_t , kurtosis-adaptive v_t^{adapt} , Student- t update, oscillator-native trust gate, and harmonic-aware frequency initialisation. This controlled comparison ensures that any performance difference is attributable solely to the noise-adaptive mechanisms.

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Author contributions statement

Y.-J.C. conceived the research framework, designed the NAROSSM algorithm, implemented the software, conducted all experiments, and wrote the manuscript. D.-Y.K. supervised the research, provided critical feedback on the methodology, and reviewed the manuscript.

Additional information

Data availability. The COHFACE dataset is publicly available at <https://www.idiap.ch/dataset/cohface>. The MAHNOB-HCI dataset is available at <https://mahnob-db.eu/hci-tagging/>. Code implementing NAROSSM and reproducing all experiments will be made available upon publication.

Competing interests. The authors declare no competing interests.